

# Mix-GEMM: An efficient HW-SW Architecture for Mixed-Precision Quantized Deep Neural Networks Inference on Edge Devices

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**Abstract**— Deep Neural Network (DNN) inference based on quantized narrow-precision integer data represents a promising research direction toward efficient deep learning computations on edge and mobile devices. On one side, recent progress of Quantization-Aware Training (QAT) frameworks aimed at improving the accuracy of extremely quantized DNNs allows achieving results close to Floating-Point 32 (FP32), and provides high flexibility concerning the data sizes selection. Unfortunately, current Central Processing Unit (CPU) architectures and Instruction Set Architectures (ISAs) targeting resource-constrained devices present limitations on the range of data sizes supported to compute DNN kernels.

This paper presents *Mix-GEMM*, a hardware-software co-designed architecture capable of efficiently computing quantized DNN convolutional kernels based on byte and sub-byte data sizes. *Mix-GEMM* accelerates General Matrix Multiplication (GEMM), representing the core kernel of DNNs, supporting all data size combinations from 8- to 2-bit, including mixed-precision computations, and featuring performance that scale with the decreasing of the computational data sizes. Our experimental evaluation, performed on representative quantized Convolutional Neural Networks (CNNs), shows that a RISC-V based edge System-on-Chip (SoC) integrating *Mix-GEMM* achieves up to 1.3 TOPS/W in energy efficiency, and up to 13.6 GOPS in throughput, gaining from 5.3× to 15.1× in performance over the OpenBLAS GEMM frameworks running on a commercial RISC-V based edge processor. By performing synthesis and Place and Route (PnR) of the enhanced SoC in Global Foundries 22nm FDX technology, we show that *Mix-GEMM* only accounts for 1% of the overall area consumption.

## I. INTRODUCTION

Deep Neural Networks (DNNs) are currently the preferred choice for artificial intelligence and computer vision tasks in both research and industrial applications. DNNs are composed of a stack of layers, whose execution time is typically dominated by the computation of large linear algebra operations like General Matrix Multiplications (GEMMs). Optimizing DNNs represents a major challenge in many fields, in particular when targeting the deployment to hardware architectures designed for edge and mobile segments, requiring high performance but presenting tight constraints in terms of area, memory, and energy consumption.

A widespread solution aimed at decreasing this burden is *quantization*, a family of techniques designed to reduce the numerical precision required to represent the parameters of a DNN and the data computed by its layers. In particular, integer quantization focuses on deploying trained DNNs with narrow-integer formats, typically ranging from 8- down to 2-bit [35], [43], [75], rather than with the standard Floating-Point 32 (FP32) data size. Quantizing DNNs exposes a large design space, as each layer can be quantized to its own precision. Moreover, input data and parameters can also be quantized differently within a layer, resulting in mixed-precision operations. Exploring this design space allows to trade-off computational requirements against quality of results, which is a key enabler of deployment in resource-constrained devices. Indeed, reducing the precision of highlighted parameters and data decreases the memory and the bandwidth required to store and load them, allowing resource-constrained devices to support larger models, or to relax constraints around the sizing of their memory hierarchy and power envelope. Quantization also enables computing operations like GEMM at low precision, with a consequent improvement in terms of performance and energy efficiency. However, in practice, most of the current general-purpose Central Processing Unit (CPU) architectures lack adequate support for efficiently handling narrow-precision formats, as most of the Instruction Set Architectures (ISAs) neither support data sizes smaller than 8-bit, nor support mixed-precision computations. Although modern Single Instruction Multiple Data (SIMD) extensions [60] and hardware accelerators [27], [52] are increasing their support for narrow-precision data sizes and mixed-precision computations on CPU architectures, they only consider a small subset of data sizes granularities. As a result, exploiting fine-grained quantization of DNNs on modern processors does not always provide a real benefit to the actual computation performance, as quantized data have to be either saved in memory in a sub-optimal format (*i.e.*, with data sizes supported by the processor ISA), or decompressed before the actual computation exploiting costly bit-manipulation operations. Therefore, investigating hardware and software architectures